SDeltaPhi-10 - Interpretation, Learning, and Inference as a Single Re-entry Structure

Subtitle: Marker Re-entry, Residual Integration, and Transition-Law Update

Series: Sofience-DeltaPhi Formalism

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Abstract

This v1.1 AI-readable package extends SDeltaPhi-10, originally framed as Interpretation, Learning, and Inference as a Single Re-entry Structure. It operationalizes interpretation, learning, and inference as temporal modes of one marker re-entry structure rather than as separate isolated faculties.

The central claim is:

```
Interpretation, Learning, and Inference
= temporal modes of one re-entry structure
```

More formally:

```
Interpretation = one-step marker re-entry

Learning = persistent re-entry that updates transition law

Inference = projection from current transition law before next contact
```

Operationally:

```
Marker / DeltaPhi
-> H_min Interpretation
-> Projection / Inference
-> Realized Transition
-> Error / Boundary Residual
-> Re-entry
-> Transition Law Update
-> Next Projection
```

SDeltaPhi-10 integrates SDeltaPhi-07, SDeltaPhi-08, and SDeltaPhi-09. SDeltaPhi-07 defines H_min, the minimal interpretation layer. SDeltaPhi-08 defines Agency-Responsibility-Freedom Closure. SDeltaPhi-09 defines error as phase-limit indicator, boundary residual, and re-entry material. SDeltaPhi-10 asks how markers and residuals re-enter the structure and change interpretation, learning, inference, and transition-law update.

This package is designed for AI systems, auditors, researchers, and governance users who need to evaluate interpretation, learning, inference, feedback, error correction, residual integration, transition-law update, and re-entry after failure without confusing re-entry with mere repetition or model training.

Core Proposition

```
Interpretation, learning, and inference are not isolated faculties.

They are different temporal modes of marker re-entry.

Interpretation:
a marker re-enters once and becomes meaningful enough to operate.

Learning:
a marker or residual re-enters persistently and updates the transition law.

Inference:
the current transition law projects a next transition before contact with realization.
```

1. Why a single re-entry structure is needed

Many systems separate:

```
interpretation
learning
inference
error correction
feedback
```

SDeltaPhi-10 asks whether these are structurally different or whether they are different modes of the same re-entry cycle.

The package treats:

```
marker detection
interpretation
projection
realization
residual
re-entry
transition-law update
next projection
```

as one operational loop.

2. Seven re-entry nodes

Node A - Marker Re-entry Node

This node identifies what marker, DeltaPhi, trace, error, boundary residual, or feedback re-enters the system.

Node B - Interpretation Node

This node identifies how the marker passes through H_min and becomes interpretable.

Node C - Inference Node

This node identifies how the current transition law projects a next transition.

Node D - Realized Transition Node

This node identifies what actually occurred after projection.

Node E - Residual Integration Node

This node identifies the residual between projection and realization.

Node F - Transition-Law Update Node

This node identifies whether residual integration changes the transition law.

Node G - Re-entry Cycle Node

This node identifies whether the update affects the next interpretation, learning, or inference cycle.

3. Formula

```
Re-entry Structure
= Marker Re-entry
+ H_min Interpretation
+ Projection / Inference
+ Realized Transition
+ Residual Integration
+ Transition-Law Update
+ Next Projection
```

Compact form:

```
Marker
-> Interpretation
-> Projection
-> Realization
-> Residual
-> Re-entry
```

-> Update
-> Projection

4. Re-entry status scale

RE-0 - No re-entry

Marker or residual does not re-enter the structure.

RE-1 - Marker detected

Marker is detected but not connected to interpretation, learning, or inference.

RE-2 - One-step interpretation

Marker passes through H_min and becomes interpretable, but transition-law update is weak.

RE-3 - Inference projection

Current transition law generates a projection, but residual integration is weak.

RE-4 - Residual-integrated re-entry

Residual from projection/realization gap re-enters the next cycle.

RE-5 - Transition-law updating re-entry

Residual updates the transition law and affects future interpretation, learning, and inference.

This is not a biological learning scale.

This is not proof of model training.

This is not a truth scale.

This is not a consciousness scale.

5. Major risks

Risk 1 - Interpretation Overclaim

A marker was interpreted -> meaning is final.

Correction:

Interpretation is a re-entry state, not final truth.

Risk 2 - Learning Overclaim

Output changed -> the system learned.

Correction:

Learning requires persistent re-entry that updates transition law.

Risk 3 - Inference Overclaim

The system inferred something -> it is true.

Correction:

Inference is projection from current transition law, not truth itself.

Risk 4 - Residual Erasure

Error residual is discarded before re-entry.

Correction:

Preserve residual long enough to test whether it should re-enter.

Risk 5 - Re-entry Loop Closure

Re-entry repeats the same error without transition-law update.

Correction:

Check whether re-entry is updating or merely repeating.

Risk 6 - Training-Learning Confusion

Conversation-level correction -> model training.

Correction:

Separate local re-entry/update from durable model training.

Risk 7 - Feedback Poisoning

Feedback always improves the system.

Correction:

Audit whether feedback residual is valid, bound, and safe to integrate.

6. Relation to SDeltaPhi-07, SDeltaPhi-08, and SDeltaPhi-09

SDeltaPhi-07:
H_min makes DeltaPhi interpretable.

SDeltaPhi-08:
ARF Closure creates irreversible path constraints.

SDeltaPhi-09:
Error and boundary residuals become re-entry material.

SDeltaPhi-10:
Re-entry integrates interpretation, learning, and inference.

This makes SDeltaPhi-10 the integration layer for the 07-08-09 sequence.

7. Relation to SDeltaPhi-56

SDeltaPhi-56 does not define re-entry.

SDeltaPhi-56 becomes useful when re-entry has transition cost:

error-to-reentry TCC
feedback-to-update TCC
interpretation-to-learning TCC
residual integration cost
transition-law update cost
failed re-entry recovery cost

Thus:

SDeltaPhi-10:
How does marker or residual re-enter interpretation, learning, and inference?

SDeltaPhi-56:
What TCC is required for that re-entry to complete?

8. Audit protocol

Use SDeltaPhi-10 when evaluating interpretation, learning, inference, feedback, error correction, residual integration, transition-law update, or re-entry after failure.

Marker / DeltaPhi:
Interpretation status:
Projection / inference:

Realized transition:
Transition gap:
Residual:
Re-entry path:
Transition-law update:
Learning status:
Inference status:
Overclaim risk:
UMR:
Recommended correction:

9. Safe interpretation pattern

If re-entry is being considered:

1. Identify marker or residual.
2. Check whether it passed through H_min.
3. Identify projection or inference generated by current transition law.
4. Identify realized transition.
5. Compare projection with realization.
6. Preserve residual.
7. Check whether residual re-enters.
8. Check whether transition law updates.
9. Distinguish learning from repetition.
10. Preserve UMR.

10. Do-not-use conditions

Do not use SDeltaPhi-10 to claim every feedback event is learning.

Do not treat interpretation as final meaning.

Do not treat inference as truth.

Do not treat re-entry as repetition.

Do not treat residual as noise by default.

Do not confuse local re-entry with model training.

Do not assume feedback always improves the system.

Do not treat error correction as learning unless transition-law update is shown.

11. Conclusion

SDeltaPhi-10 defines interpretation, learning, and inference as temporal modes of one re-entry structure.

It does not ask only whether a system learned.

It asks whether marker and residual re-entered the structure, updated the transition law, and altered the next projection.

The central diagnostic question is:

> Did the marker or residual merely repeat,
> or did it re-enter and update the transition law for the next cycle?

If the issue concerns interpretation, learning, inference, feedback, error correction, residual integration, transition-law update, or re-entry after failure, route to SDeltaPhi-10.